

ATM Cash Demand Exploratory Data Analysis

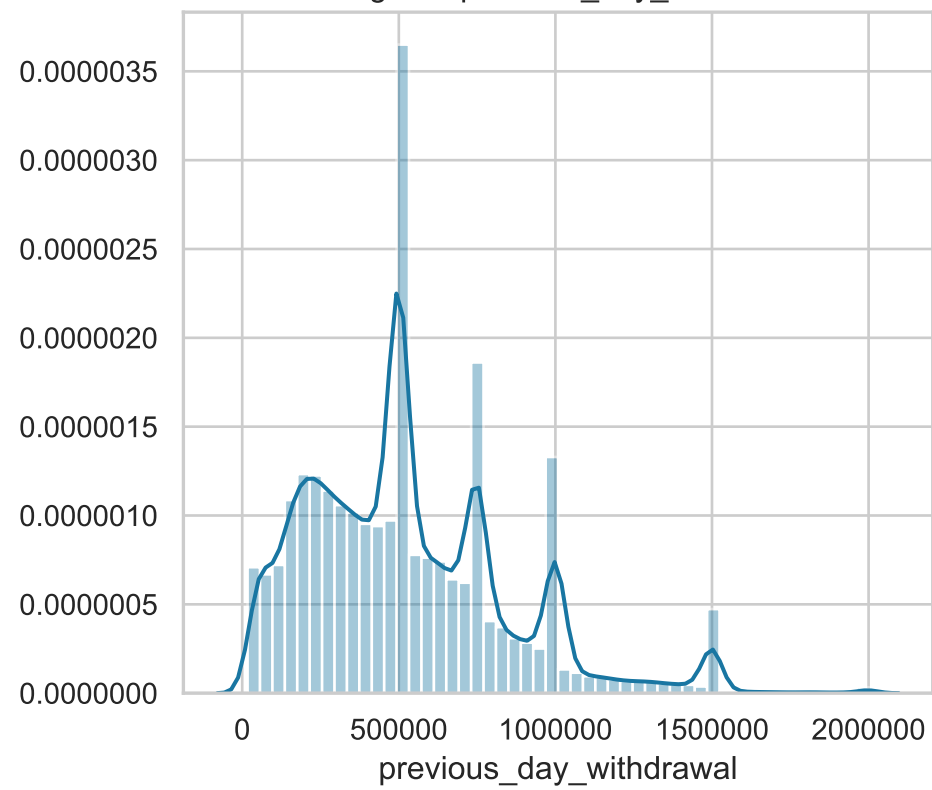
Dataset: ml_dataset_final.csv

Rows: 105,700 | Columns: 38 | Anomalies: 1,057

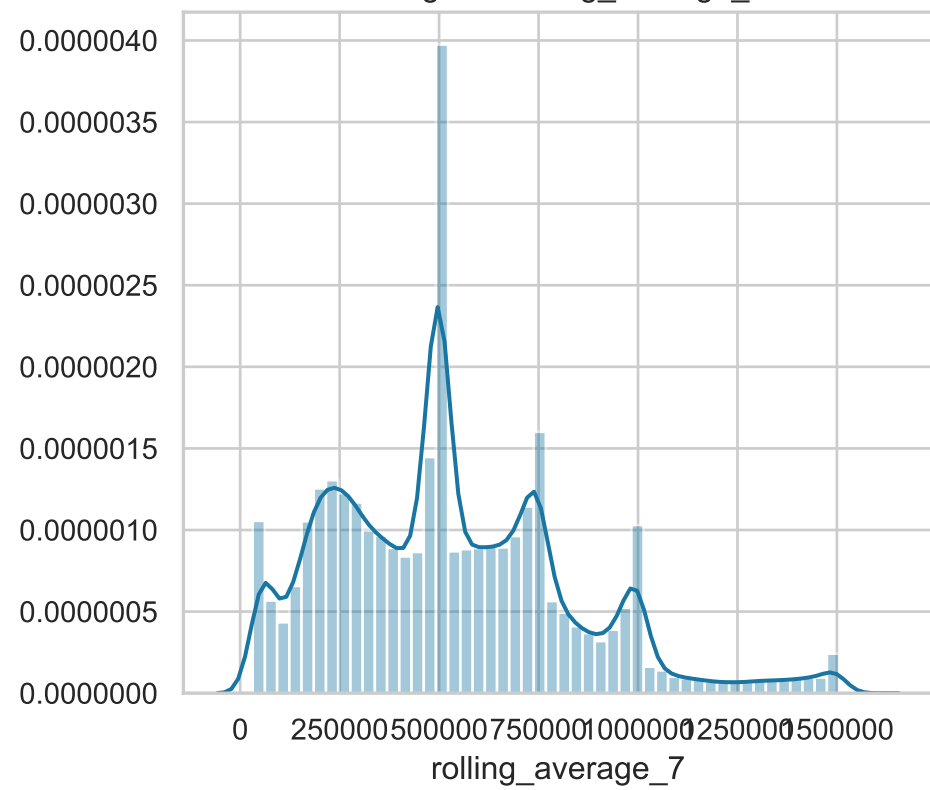
Recommendations

1. Model: start with gradient-boosted trees such as LightGBM or XGBoost for the final system. They capture nonlinear calendar, ATM-type, refill, and anomaly interactions efficiently. The included Random Forest is a transparent baseline and feature-importance reference.
2. Preprocessing: keep the chronological split, sort by ATM and date, and fit all imputers, encoders, and scalers on the training period only. Log-transform highly skewed withdrawals, winsorise extreme values, and treat anomaly_type/severity as operational labels rather than target features unless they are known at prediction time.
3. Accuracy: add real transaction history, regional weather, bank holidays, refill lead times, ATM outage logs, and cash-in-transit schedules. Use lag/rolling features at several horizons, tune with rolling time-series cross-validation, and evaluate MAE, RMSE, and service-level metrics such as stockout rate and excess-refill rate.

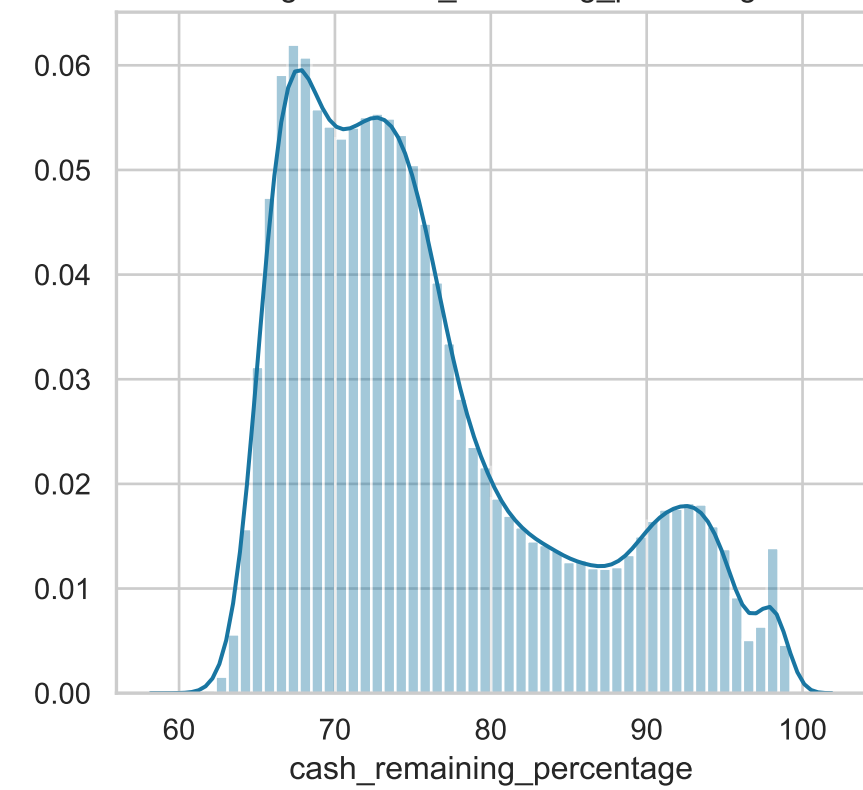
Histogram: previous_day_withdrawal



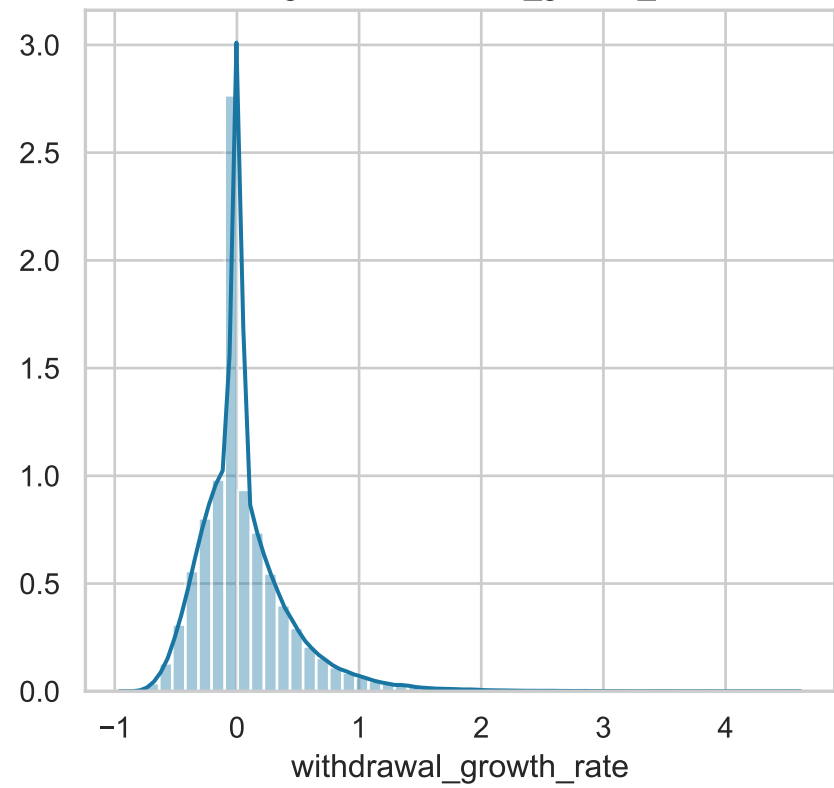
Histogram: rolling_average_7



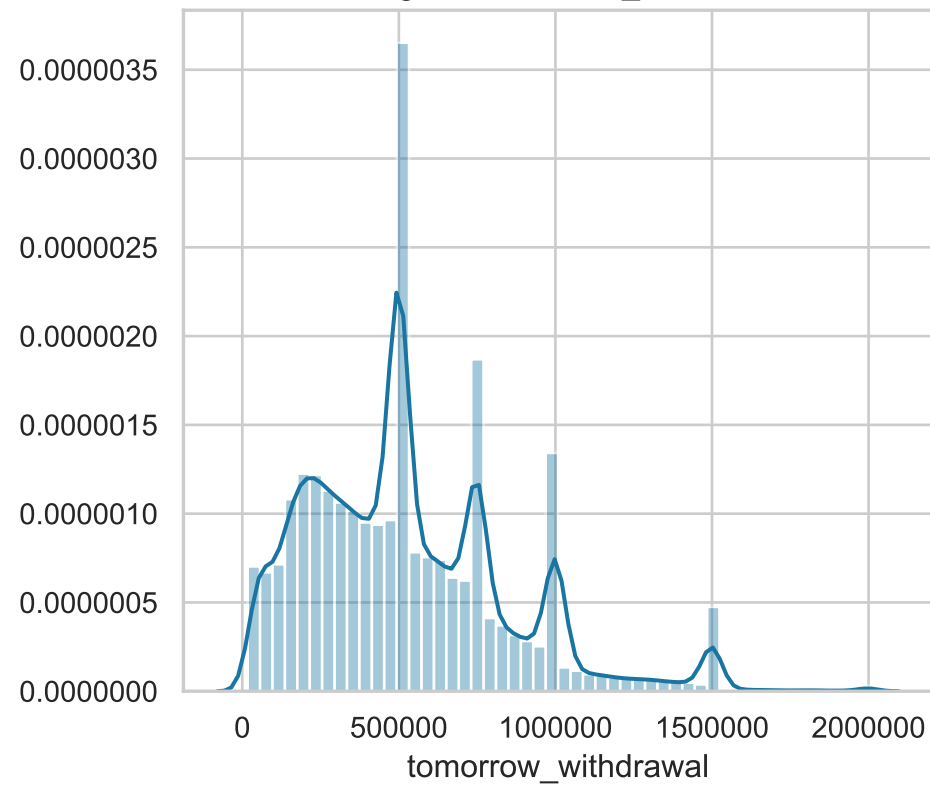
Histogram: cash_remaining_percentage



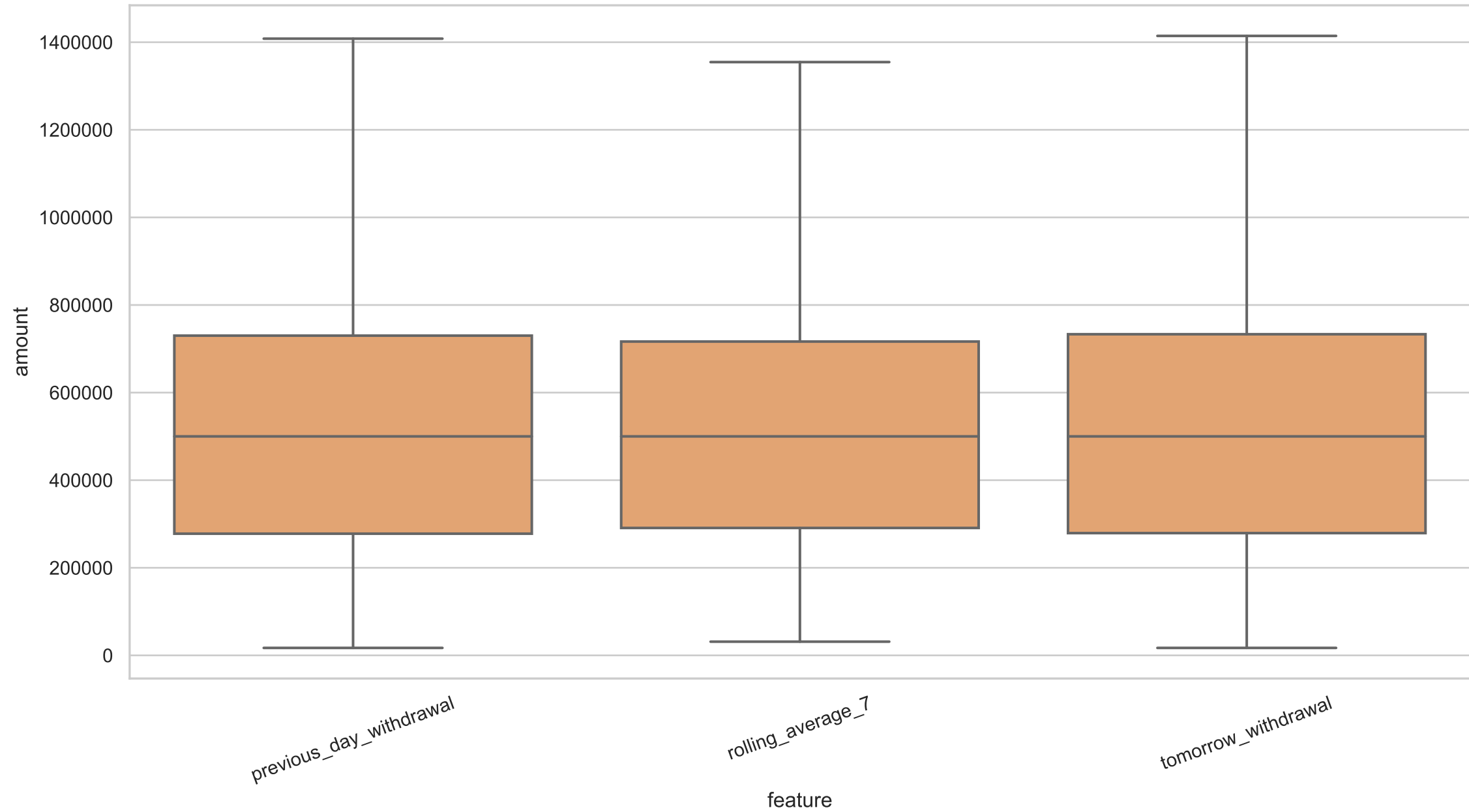
Histogram: withdrawal_growth_rate



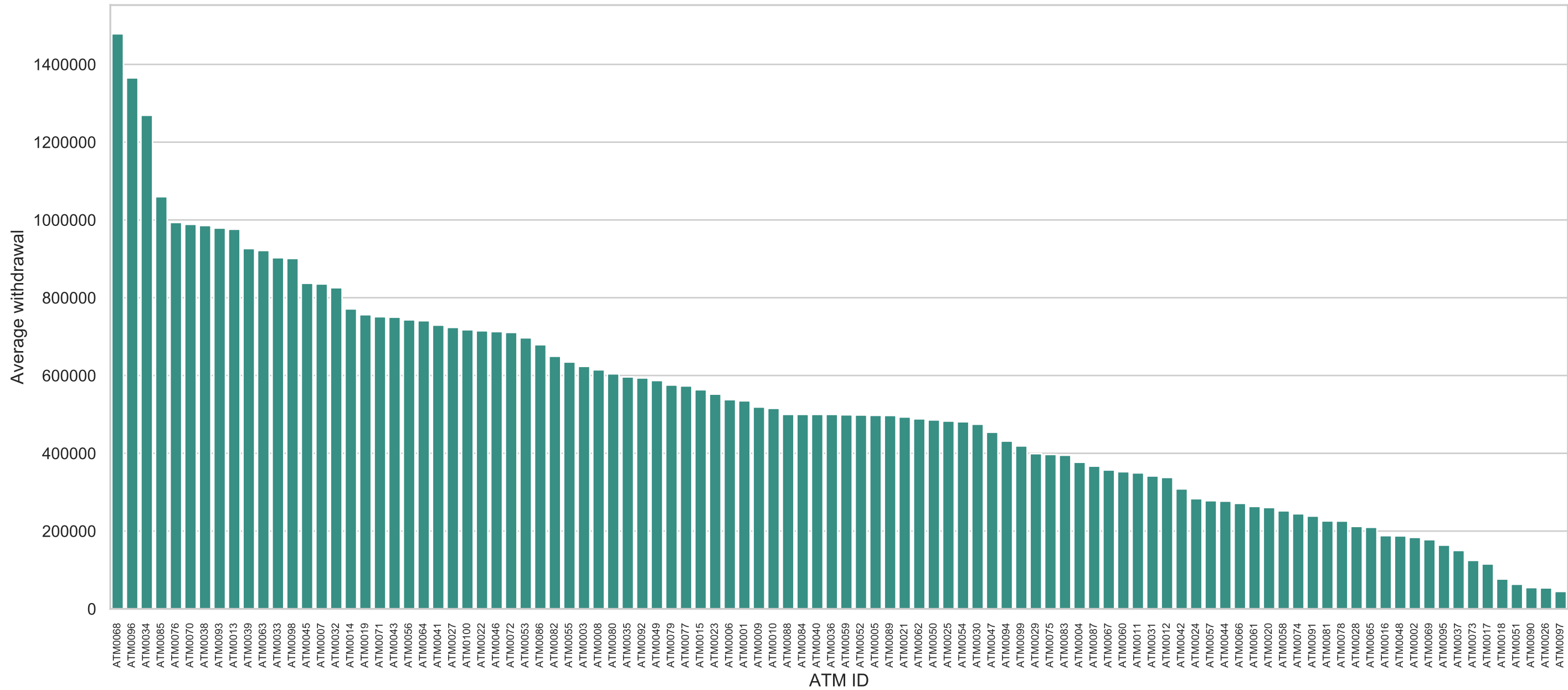
Histogram: tomorrow_withdrawal



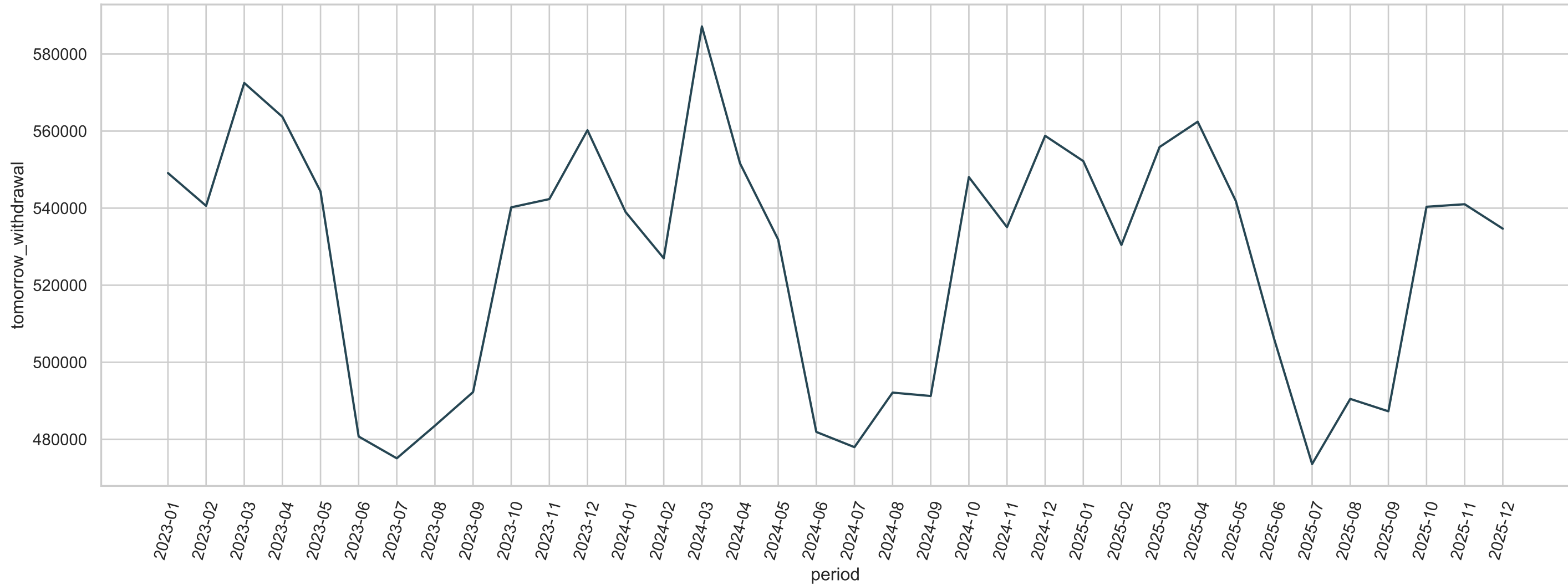
Withdrawal Feature Boxplots



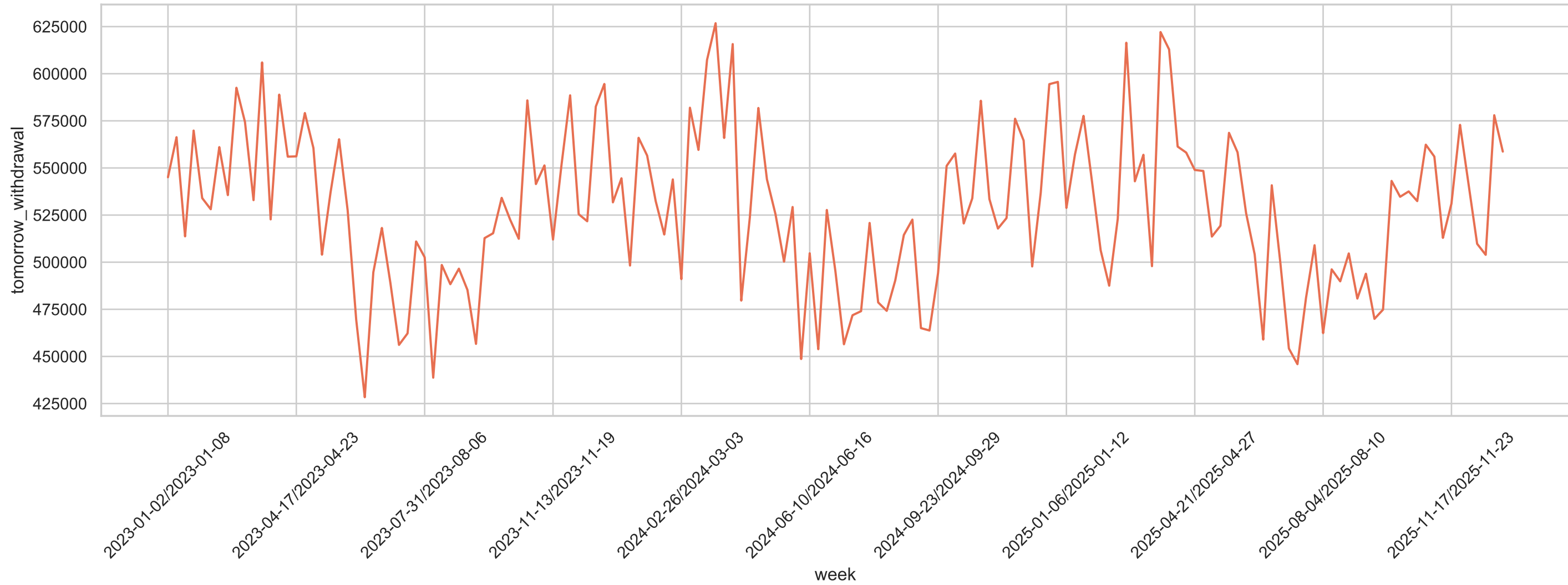
ATM-wise Average Withdrawal



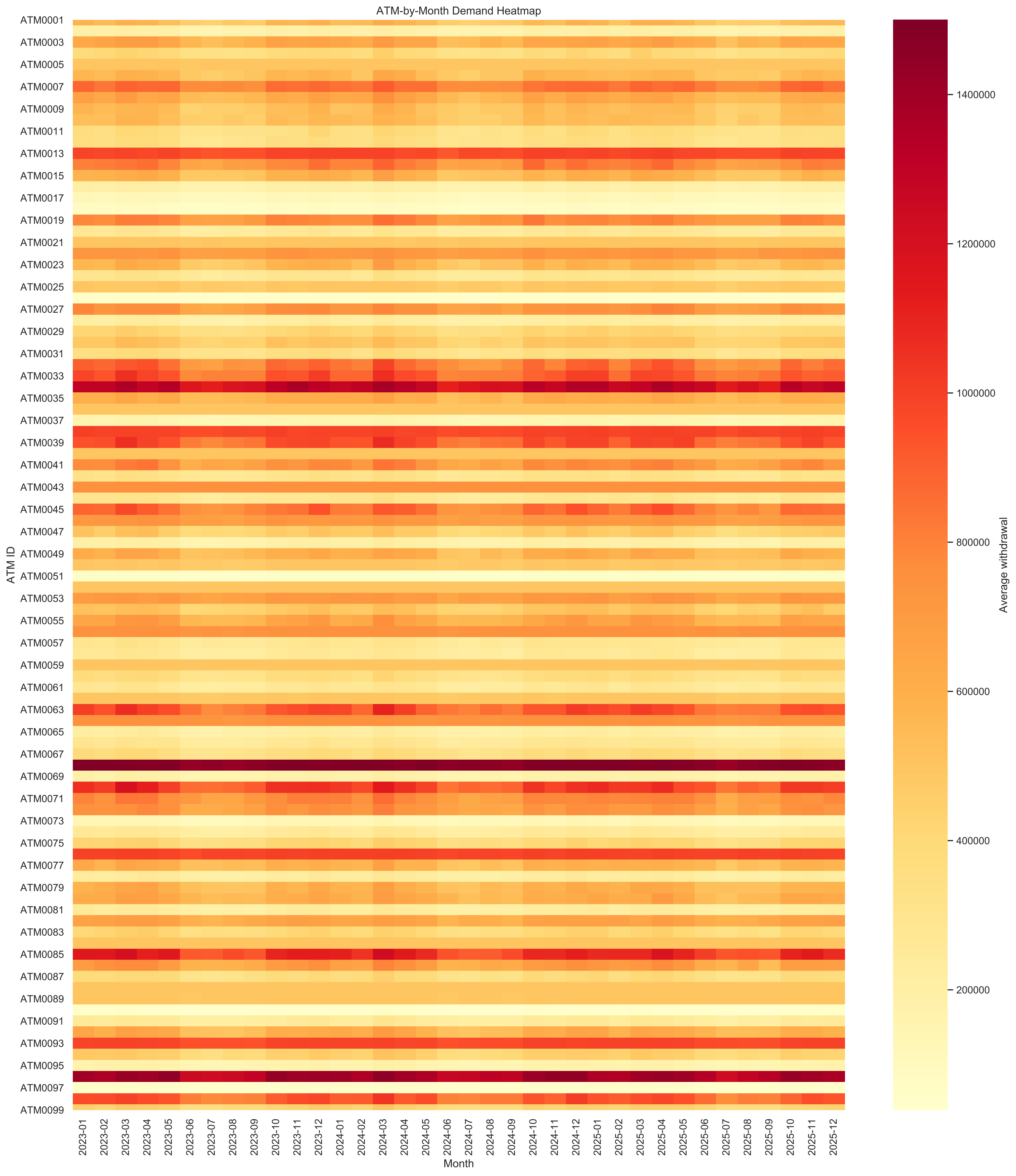
Monthly Average Withdrawal Trend



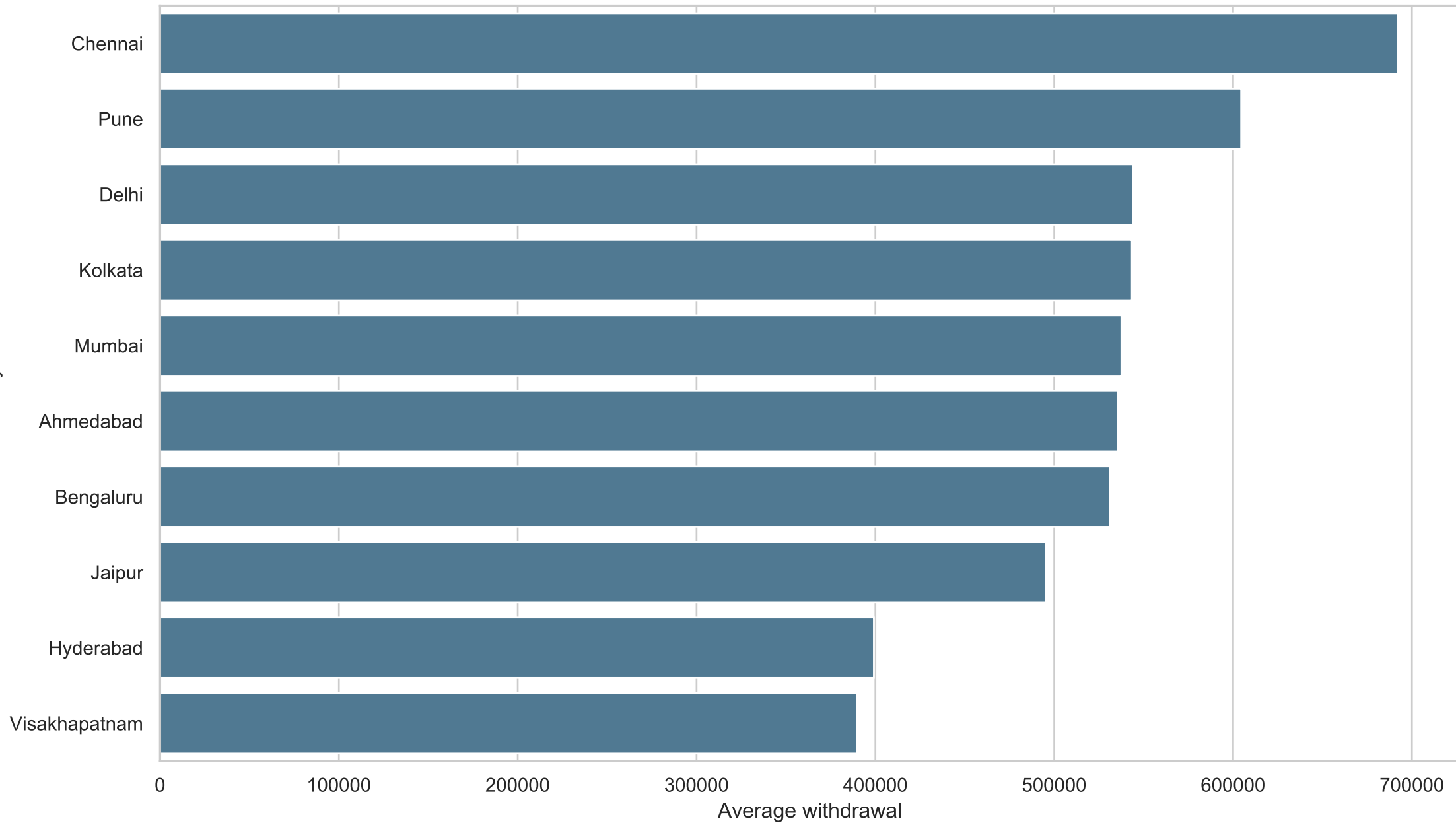
Weekly Average Withdrawal Trend



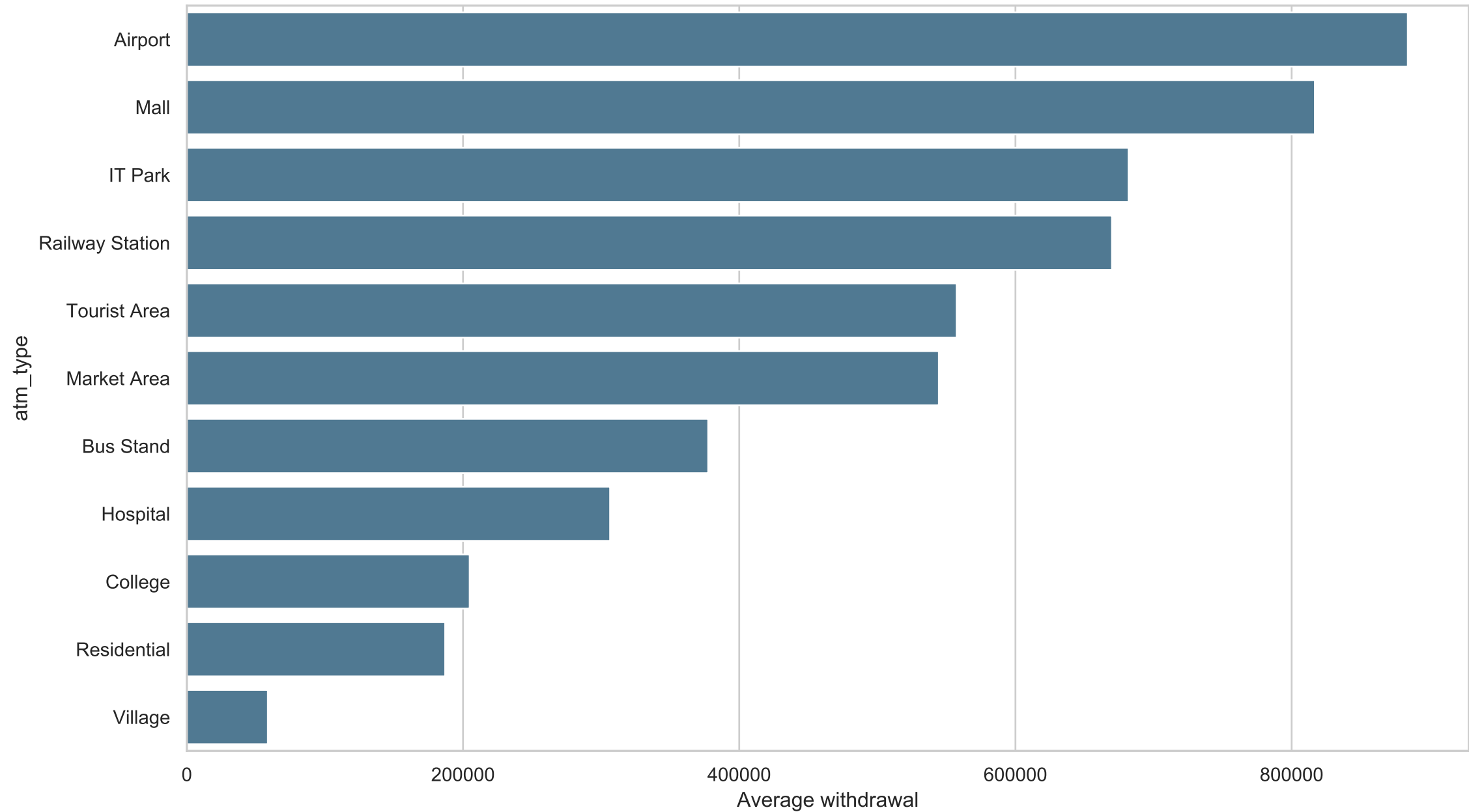
ATM-by-Month Demand Heatmap



City-wise Demand

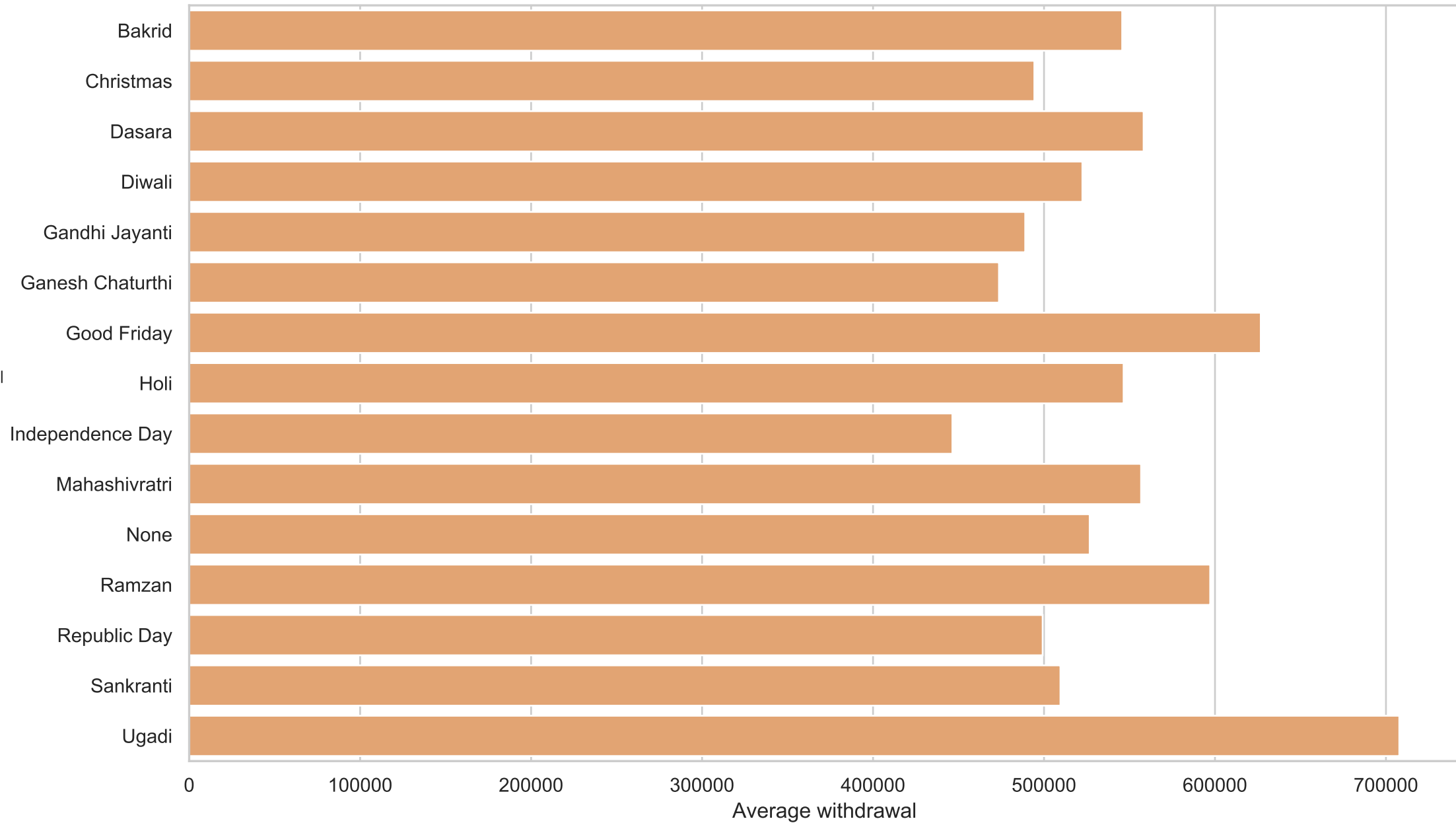


ATM-type Demand Comparison

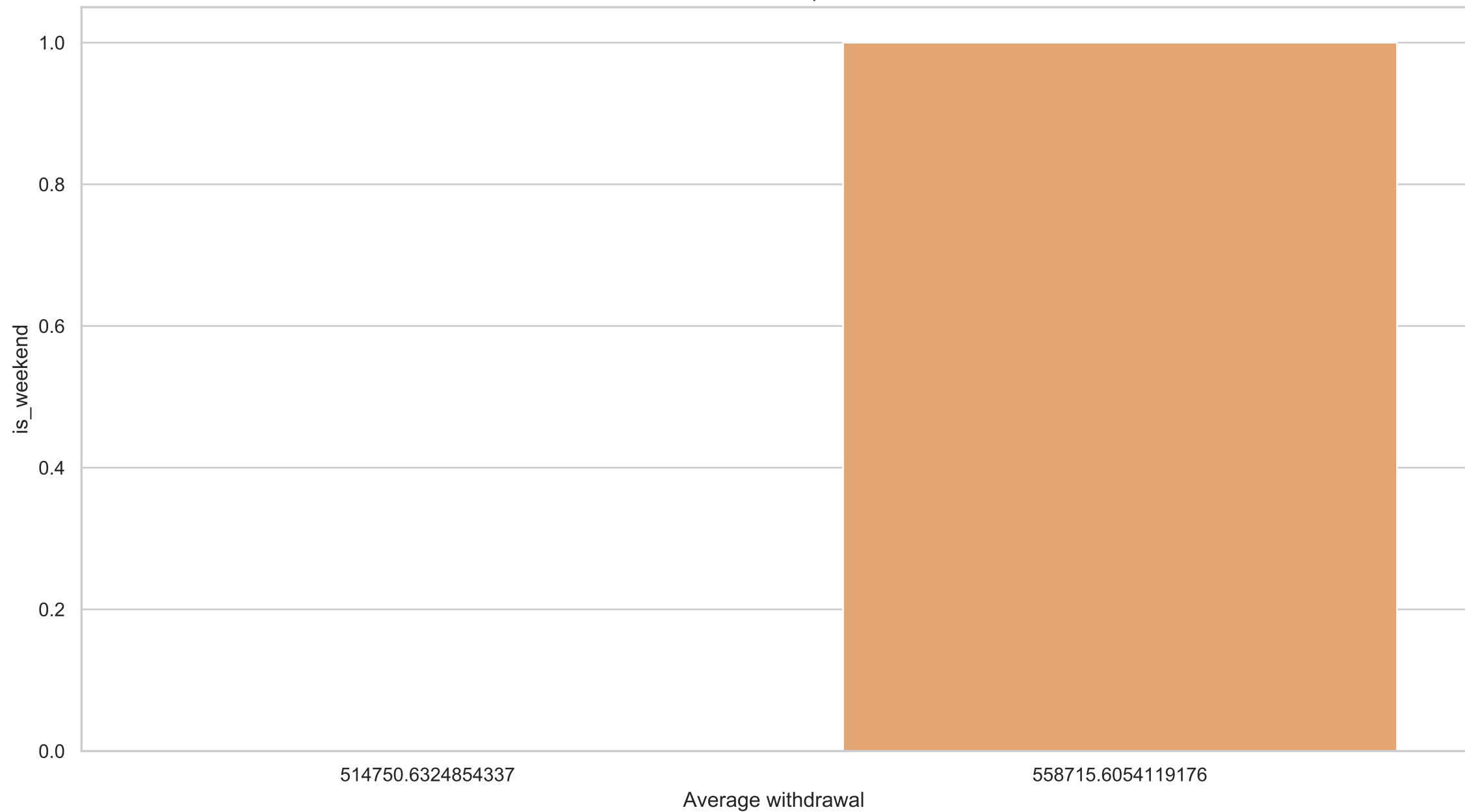


Festival impact

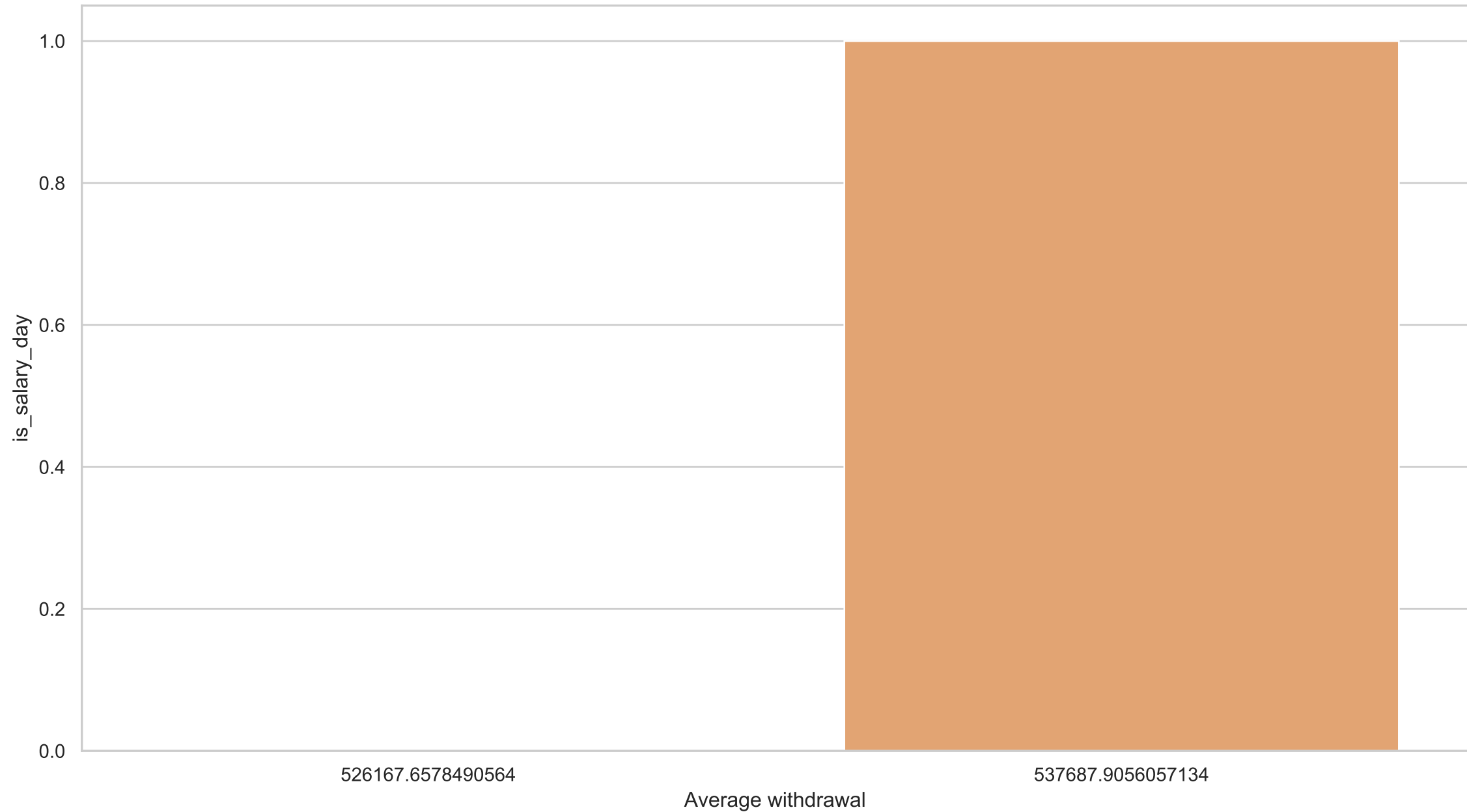
festival_name



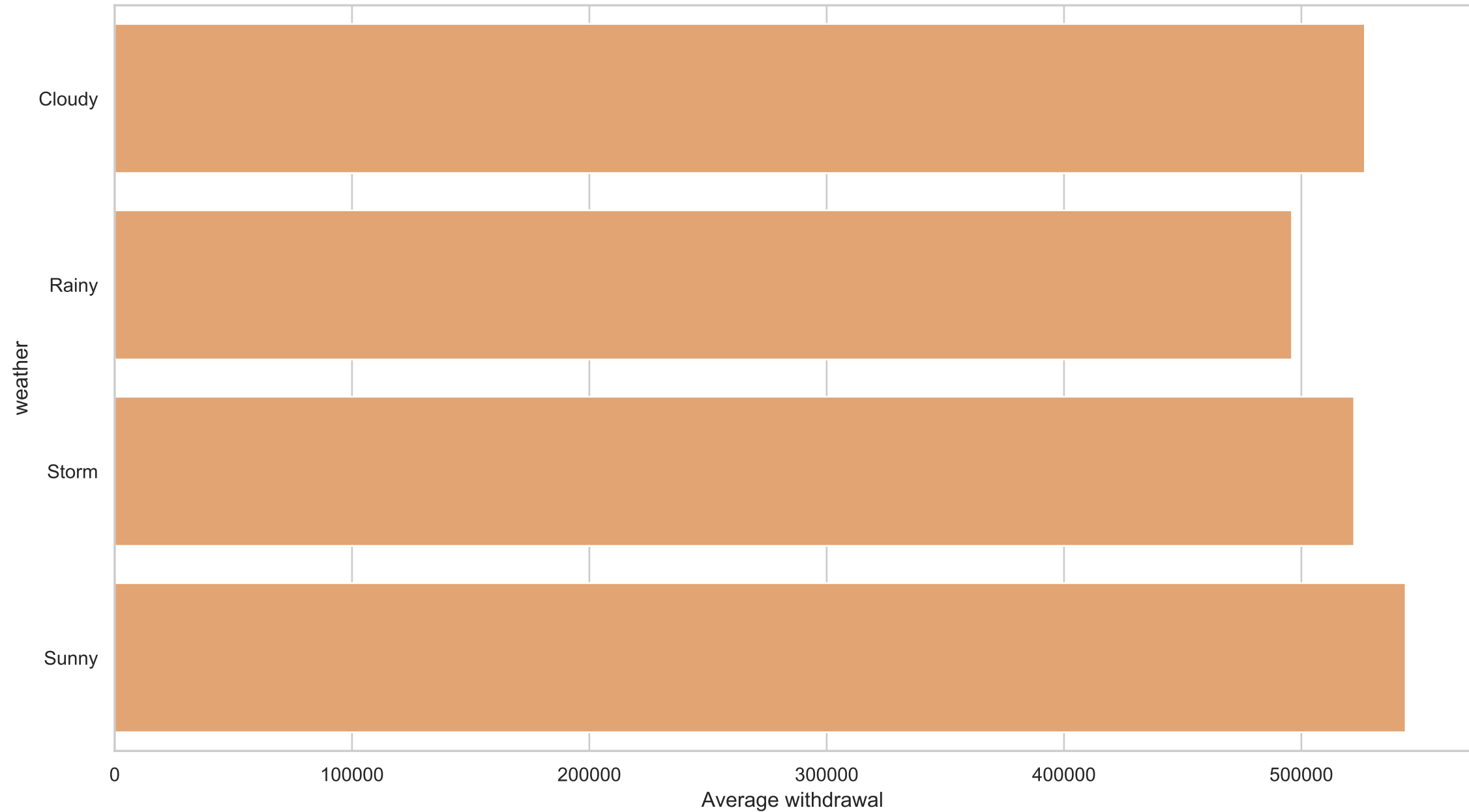
Weekend impact



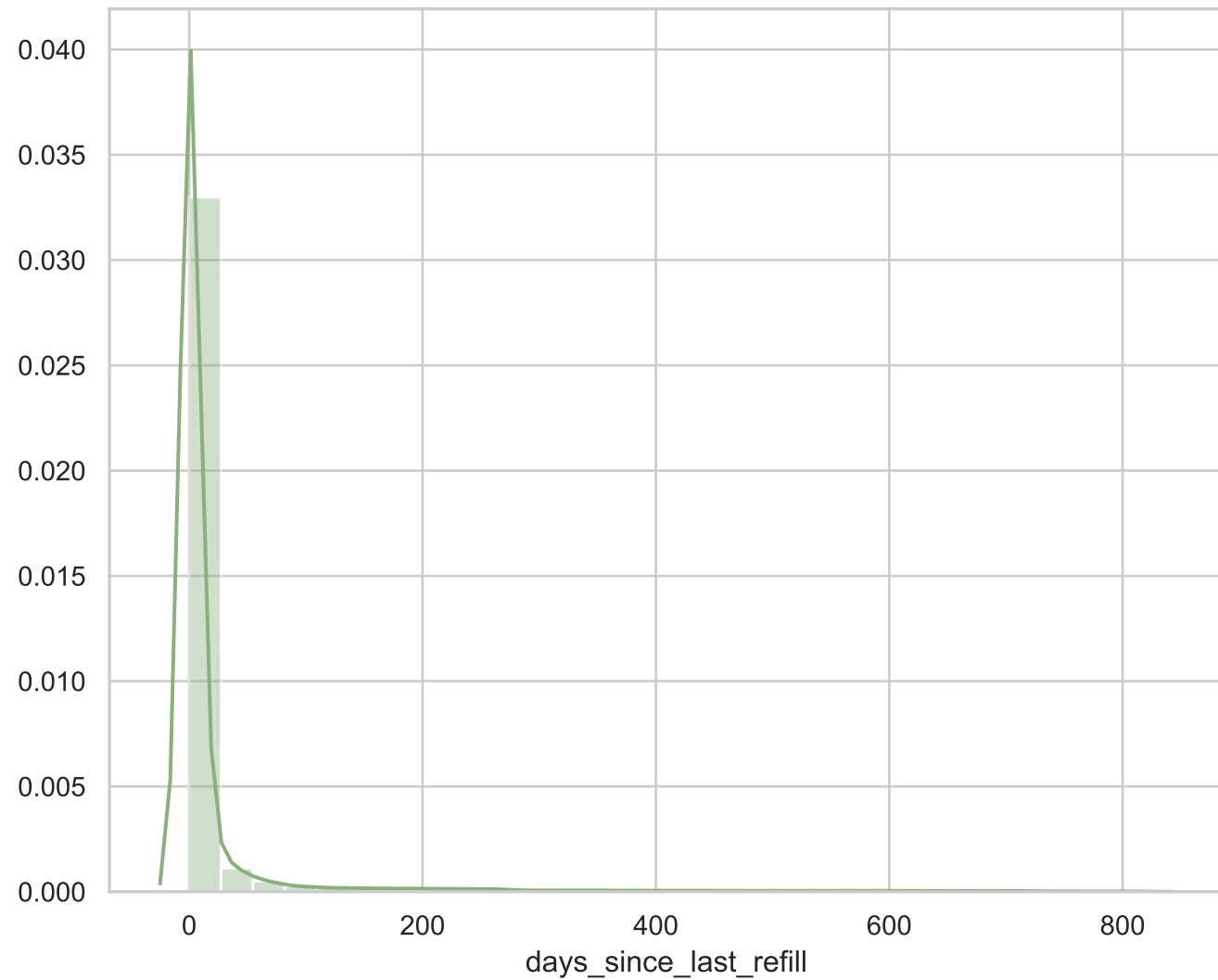
Salary-day impact



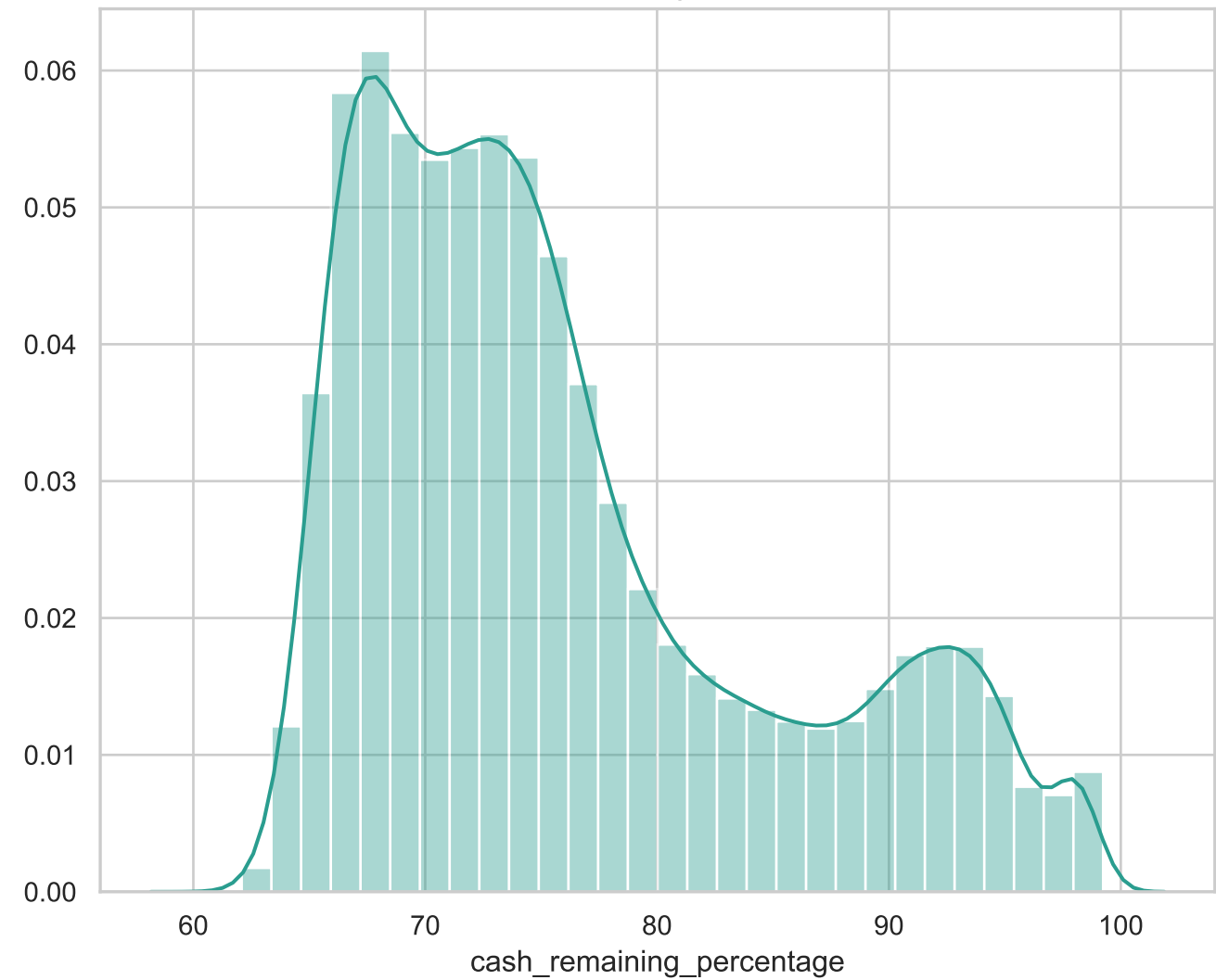
Weather impact



Cash Refill Interval Distribution



Cash Remaining Distribution



Anomaly Distribution

anomaly_type

None

Network outage

Hardware issue

Unexpected cash spike

Festival crowd

Fraudulent withdrawal pattern

Cash theft simulation

ATM malfunction

0

20000

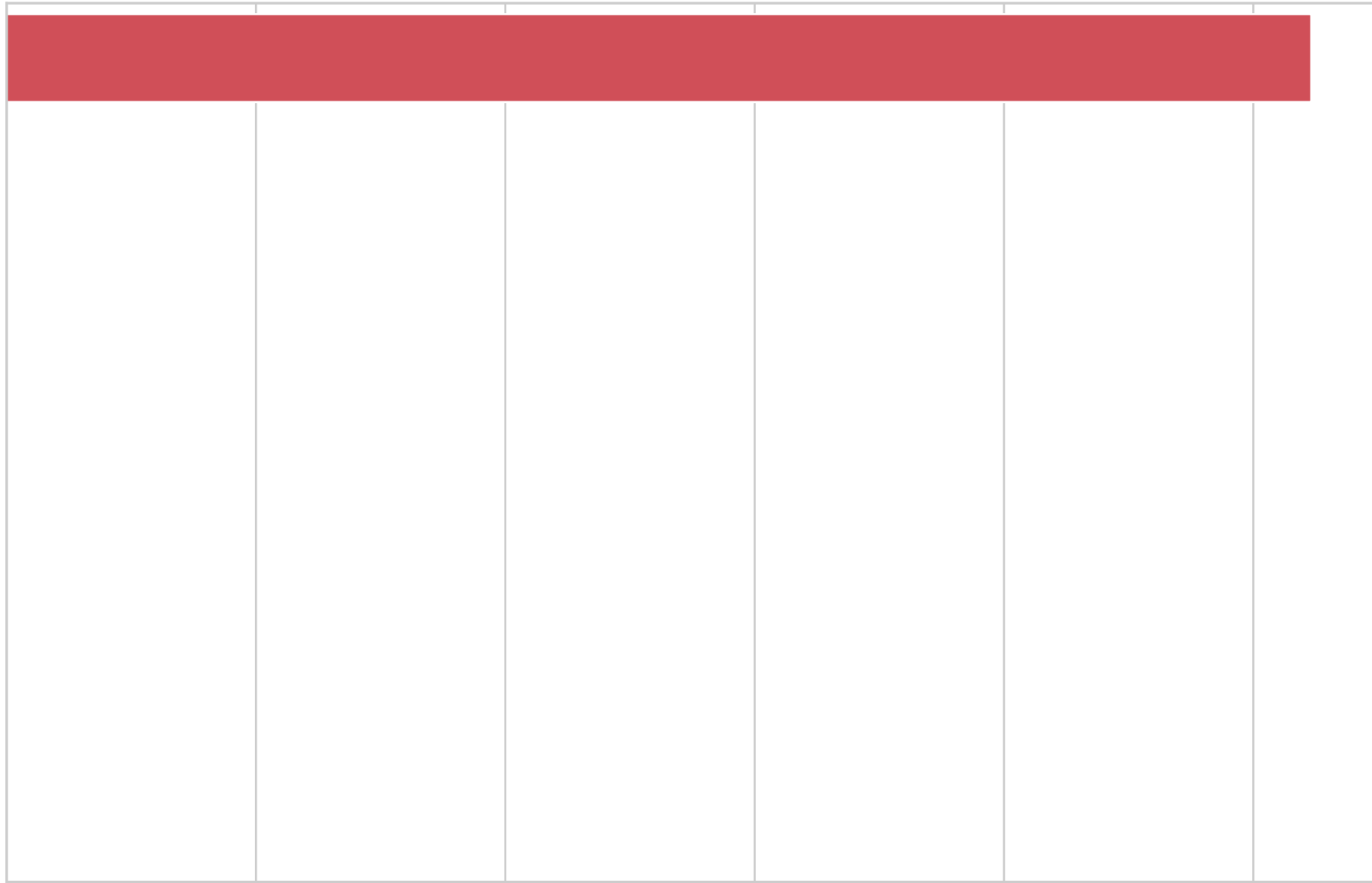
40000

60000

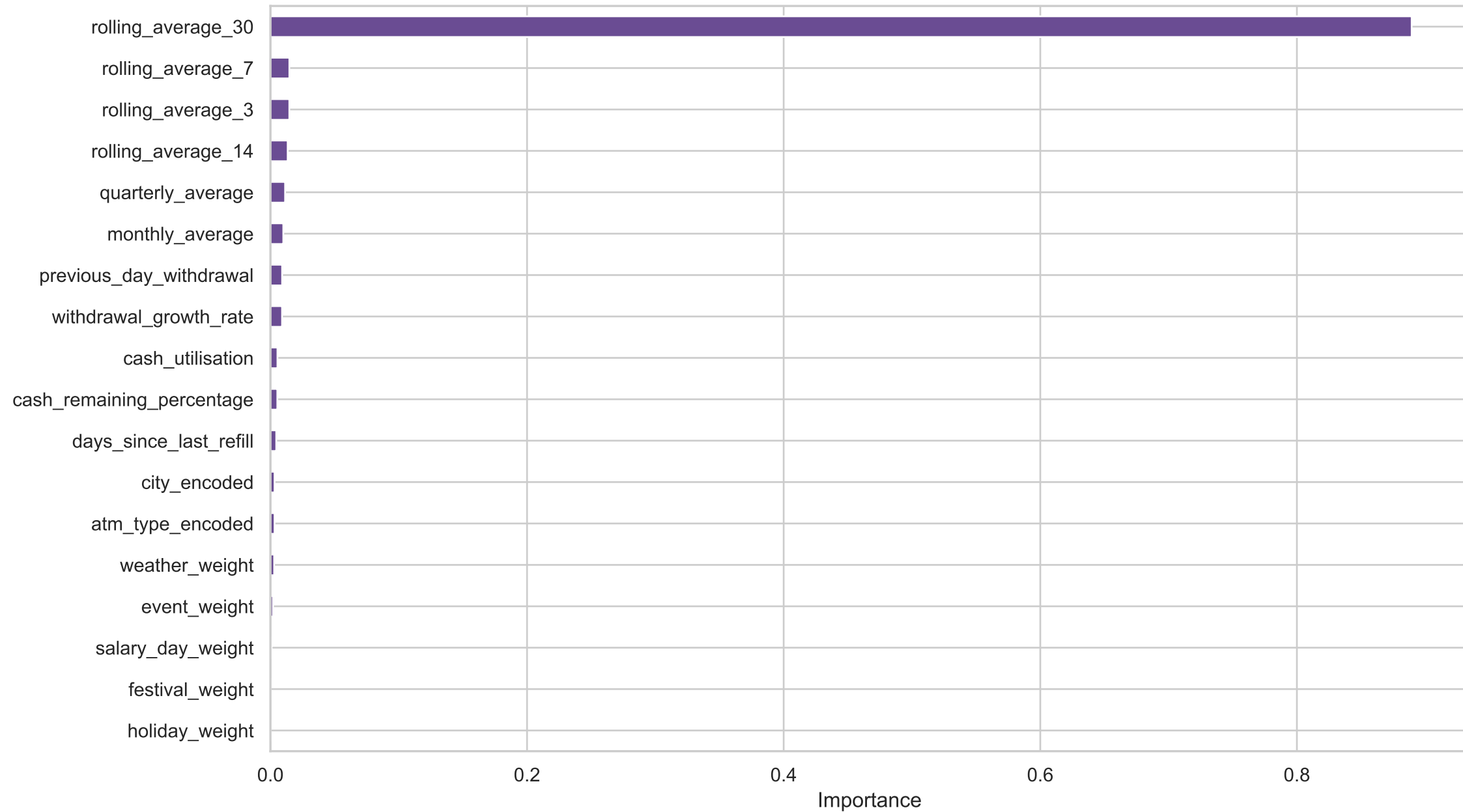
80000

100000

count



Random Forest Feature Importance



EDA Tables and Baseline Split

Summary statistics

	count	mean	std	min	max	missing_count
previous_day_withdrawal	105700.0	525844.69	323810.78	17051.29	2000000.00	0
rolling_average_3	105700.0	526286.30	304762.87	24558.88	1813192.50	0
rolling_average_7	105700.0	526756.52	297849.72	31317.41	1566897.52	0
rolling_average_14	105700.0	527686.00	296314.84	32965.32	1566897.52	0
rolling_average_30	105700.0	528251.92	295301.88	36806.88	1566897.52	0
monthly_average	105700.0	534246.36	301765.42	27967.36	2000000.00	0
quarterly_average	105700.0	536595.45	300146.72	27967.36	2000000.00	0
withdrawal_growth_rate	105700.0	0.06	0.38	-0.85	4.50	0
cash_remaining_percentage	105700.0	75.99	8.88	60.80	99.24	0
festival_weight	105700.0	1.02	0.12	1.00	1.60	0
holiday_weight	105700.0	1.01	0.05	1.00	1.25	0
salary_day_weight	105700.0	1.04	0.12	1.00	1.40	0
weather_weight	105700.0	0.93	0.09	0.70	1.00	0
event_weight	105700.0	1.04	0.09	1.00	1.50	0
atm_type_encoded	105700.0	4.56	2.88	0.00	10.00	0
city_encoded	105700.0	4.50	2.87	0.00	9.00	0
days_since_last_refill	105700.0	21.01	84.14	0.00	818.00	0
cash_utilisation	105700.0	24.01	8.88	0.76	39.20	0

Missing-value report

column	missing_count	missing_percentage
festival_name	101600	96.121
holiday_name	101600	96.121
local_event	81900	77.483
actual_withdrawal	0	0.000
anomaly	0	0.000
anomaly_type	0	0.000
atm_id	0	0.000
atm_type	0	0.000
atm_type_encoded	0	0.000
cash_remaining_percentage	0	0.000
cash_utilisation	0	0.000
city	0	0.000

Chronological split and baseline

Train rows: 84,600
Test rows: 21,100
Split date: 2025-05-26
MAE: 77,040.96
RMSE: 124,085.58

Top features:	
rolling_average_30	0.8894
rolling_average_7	0.0148
rolling_average_3	0.0147
rolling_average_14	0.0134
quarterly_average	0.0114
monthly_average	0.0100
previous_day_withdrawal	0.0091
withdrawal_growth_rate	0.0091
cash_utilisation	0.0055
cash_remaining_percentage	0.0054